

HW6: Optimal FIR and Short-time Fourier Transform

Due on Sunday June 14, 2026

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1. DFT and FFT (30%).

(a) Let $x[n] = \sin \frac{\pi n}{2}$, for $0 \leq n \leq 7$. Calculate the 8-point DFT for $x[n]$; that

is, find $X[k] = \sum_{n=0}^7 x[n] e^{-\frac{jk(2\pi)}{8}n}$, for $k = 0, 1, \dots, 7$.

(b) Estimate how many multiplications is necessary if we compute $X[k]$ for $k = 0, 1, \dots, 7$ using FFT.

(c) In the PPT slides, we show an architecture of 16-point FFT, and the frequency bins on the right seems to be in a random order: $\{0, 8, 4, 12, 2, 10, \dots, 7, 15\}$. Find out how this order is determined.

2. Information retrieval: Dual-tone multiple frequency (40%). Run `<DSP2025_HW5_DualTones.m>`. Note that the spectrogram is presented in 3D so you can drag it to change the viewing angle.

(a) [20 pts] Change the window length and the window type and explain how the result differs. Find out how short we can set the window length while still seeing two separated peaks, using each window type (Hann, Rectangular, or Blackman)?

(b) [10 pts] Load `<unknown_phonenum.wav>` and figure out what the phone number is. Explain how you obtain your answer.

3. Optimal FIR design (30%). Run `<myTestPM.m>` (which uses `<myChebyPol.m>`) to see how the Parks-McClellan algorithm iteratively approaches an optimal design of low-pass filter with equal ripple.

(a) Change the initial guess of alternating points (line 10) and discuss how your guess affects the number of iteration steps it takes.

(b) Change the cut-off frequency ω_c and discuss how it affects the final result of maximal approximation error (i.e., the ripple size).

(c) Note that the delta value *increases* by each iteration. Can you explain it?